Smart Warehouse Monitoring System Integrating ESP-32 and RFID Identification

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Abstract—Warehouse operations depend on conditions that are easy to state and hard to observe continuously: goods must be held within a temperature and humidity band, unauthorized movement must be noticed, stock and personnel movements must be identified, and storage space must be accounted for. These functions are usually handled by separate systems or by manual inspection, which is intermittent by nature. This paper presents a prototype warehouse monitoring node built around a single ESP32 microcontroller, integrating four sensing modalities and two output channels. A DHT22 provides temperature and relative humidity, a passive infrared sensor detects motion in the monitored aisle, an HC-SR04 ultrasonic transducer measures clearance distance for storage occupancy, and an MFRC522 reader identifies 13.56 MHz RFID cards presented at the node. Threshold violations raise local alerts through a buzzer and an indicator LED, while a 16x2 I2C liquid crystal display shows live readings and scan events at the point of installation, and Wi-Fi connectivity carries readings to a cloud dashboard for remote visualization. The node was developed in the Wokwi simulation environment and then realized as a physical breadboard prototype, allowing the firmware to be tested against deterministic stimuli before hardware integration. We describe the hardware interfacing, the polling firmware architecture, the alert logic, and the test methodology, and we analyze the design limitations that a production deployment would need to address.

Keywords—Internet of Things, ESP32, warehouse monitoring, DHT22, PIR motion detection, HC-SR04 ultrasonic ranging, MFRC522 RFID, embedded firmware, cloud dashboard

I. INTRODUCTION

A warehouse is a controlled environment whose control is largely assumed rather than verified. Stored goods, whether pharmaceutical, agricultural, electronic, or textile, degrade outside a temperature and humidity band, and the degradation is cumulative and usually invisible until the goods are inspected or sold. Physical security is similarly discontinuous: entry is controlled at a door, but movement inside an aisle at three in the morning is typically recorded only if a camera happened to be pointed at it and someone later looks. Stock and personnel identification is often still performed by writing on a clipboard.

In contrast to traditional warehouse surveillance methods that typically rely on a single type of sensor, the proposed SENTINEL system merges various monitoring capabilities into one ESP32-based node. This solution combines temperature and humidity tracking, PIR motion sensing, ultrasonic distance measurement, and RFID identification, offering both on-site and remote data transmission. On-site alerts are delivered via an LCD screen, buzzer, and LED, while remote access is enabled through Wi-Fi and a cloud dashboard. Consequently, this study emphasizes functional integration, cost-effective deployment, and an examination of the engineering compromises involved in housing multiple sensing and communication tasks on a single microcontroller.

Each of these problems has a mature commercial answer, but the answers arrive as separate systems with separate installation, cabling, and recurring cost. For a small or medium warehouse, the aggregate cost is frequently what prevents any of them from being deployed, and the facility continues to operate on periodic manual inspection. The falling cost of microcontrollers with integrated wireless connectivity changes the calculation, because a single inexpensive device now has enough peripherals, memory, and radio capability to host several of these functions at once.

This paper describes SENTINEL, a prototype that takes this consolidation as its design premise. All sensing, local alerting, local display, and network communication are hosted on one ESP32 module. The system was first built and exercised in the Wokwi simulator, which models the microcontroller and its peripherals and permits repeatable stimulation of sensor inputs, and was then assembled as a breadboard prototype so that the firmware could be validated against real sensor behaviour rather than against idealised models. The contributions of the paper are as follows.

- A consolidated four-modality warehouse monitoring node, combining climate sensing, motion detection, ultrasonic ranging, and RFID identification on a single microcontroller with both local and remote reporting.
- A complete hardware interfacing description, including the peripheral bus assignment and pin map, sufficient for the prototype to be reproduced.
- A polling firmware architecture with threshold-based alerting, together with an analysis of its timing behaviour and of the constraints that a blocking main loop imposes on event responsiveness.
- A simulation-first development method in which the firmware is validated in Wokwi before hardware assembly, and an assessment of what this method does and does not verify.

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- An explicit treatment of the security and reliability limitations of the prototype, including the status of RFID identification as an identifier rather than an authenticator.

The main contribution of this study is the integration of environmental, motion, distance, and RFID monitoring capabilities into a unified ESP32-based warehouse node. This setup features separate local and cloud reporting channels and includes an examination of the trade-offs involved in its implementation.

Section II reviews related work. Section III presents the system architecture and Section IV the hardware design. Section V describes the firmware, Section VI the connectivity and dashboard, and Section VII the security considerations. Sections VIII to X cover testing, limitations, and conclusions.

II. Related Work

A. Wireless Sensing for Storage and the Supply Chain

Wireless sensor networks have been applied to environmental monitoring in storage and transport for well over a decade. Ruiz-Garcia et al survey the technologies and applications in the agri-food sector and document both the value of continuous condition monitoring and the practical obstacles, including node power budgets, packet loss in cluttered environments, and measurement error induced by the operating environment itself. Broader treatments of the Internet of Things by Atzori et al and Gubbi et al place such deployments within a general architecture of sensing, connectivity, and cloud analytics that the present system follows.

B. RFID in Inventory and Access Applications

Radio frequency identification is long established for item and personnel identification; Want provides a foundational overview of RFID technology and its operating principles. Juels examines RFID security and privacy concerns, while Garcia et al. demonstrate that the proprietary Crypto1 cipher used in MIFARE Classic cards can be compromised in practice. Section VII treats this as a design constraint on what the prototype can legitimately claim.

C. Microcontroller Platforms and Simulation

The ESP32 integrates a dual-core processor, Wi-Fi, Bluetooth, and a peripheral set covering the SPI, I2C, and one-wire interfaces required by this design, which is what makes single-device consolidation feasible. Browser-based simulation of such platforms has become a practical part of the development cycle, permitting firmware logic to be exercised before components are procured and wired.

D. Positioning

Published warehouse and storage monitoring prototypes typically address one modality, most often temperature and humidity, and report it to a dashboard. The present work differs in consolidating four modalities and two independent reporting paths, local and remote, on one device, and in reporting the resulting engineering trade-offs rather than only the successful case. The system's primary innovation is the integration of various warehouse monitoring capabilities into a single node powered by an ESP32. Rather than focusing solely on environmental data or specific security and identification tasks, this approach combines temperature and humidity tracking, PIR motion detection, ultrasonic distance measurement, and RFID identification. These features operate via two separate reporting channels: local notifications using an LCD screen, buzzer, and LED, alongside remote monitoring through Wi-Fi and a cloud-based dashboard. Additionally, the development process prioritizes simulation, where firmware is first tested in Wokwi before being deployed on a physical breadboard prototype. Consequently, the system's value stems not from novel sensing technologies, but from its cost-effective consolidation and evaluation of multiple warehouse monitoring functions within one unit.

III. System Architecture

The ESP32 functions as the central processing and communication unit. The sensors are connected through interfaces suited to their operating requirements: the DHT22 uses a single-wire protocol, the LCD communicates over I2C, the MFRC522 RFID reader uses SPI, and the PIR and HC-SR04 sensors rely on digital input and output pins with software-based timing. Outputs are split into two paths: a local path with the display, buzzer, and LED that keeps working even if the network goes down, and a remote path that sends data over Wi-Fi to a cloud dashboard. This setup is intentional because network issues are common in a warehouse, so alerts need to work even when connectivity fails.

Table I gives the component set in the prototype and the pin assignment used in the prototype.

TABLE I. Components, Interfaces, and Pin Assignment

Component	Interface	ESP32 pin
ESP32 development module	Host controller	—
DHT22 temperature and humidity	Single-wire	GPIO 4
PIR motion sensor (HW-416)	Digital input	GPIO 13
Active buzzer	Digital output	GPIO 12
Indicator LED	Digital output	GPIO 14
MFRC522 RFID reader (SS)	SPI	GPIO 15
MFRC522 RFID reader (RST)	SPI	GPIO 27
HC-SR04 ultrasonic (TRIG)	Digital output	GPIO 26
HC-SR04 ultrasonic (ECHO)	Digital input	GPIO 25
16x2 LCD, address 0x27	I2C	SDA / SCL

a.

Fig. 1 shows the node under simulation, and Fig. 2 shows the assembled breadboard prototype.

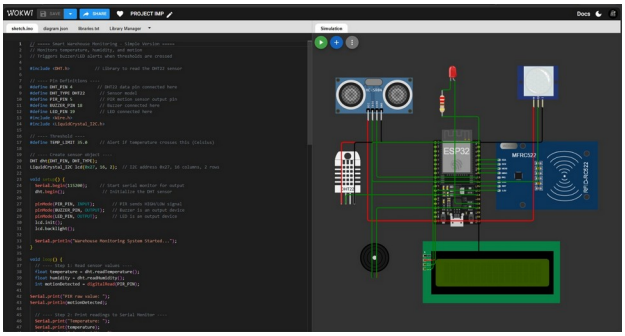


Fig. 1. The monitoring node under simulation in Wokwi, showing the firmware alongside the wired peripheral set used during development.

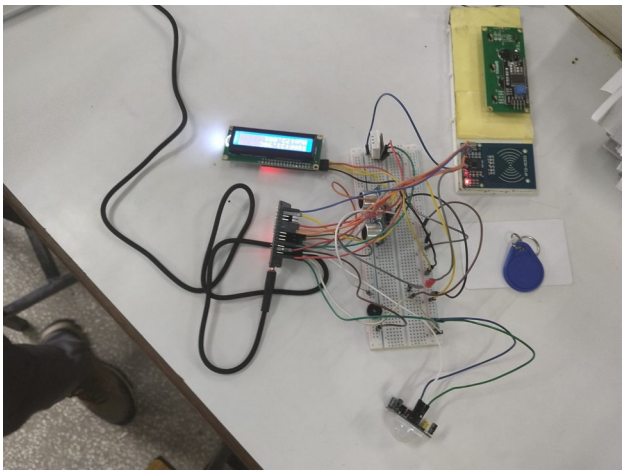


Fig. 2. The assembled breadboard prototype, with the ESP32, the character display, the ultrasonic transducer, the RFID reader with card and fob, and the motion sensor.

IV. Hardware Design and Interfacing

A. Climate Sensing

The DHT22 was chosen over the more economical DHT11 because its broader measurement range and higher resolution are essential for detecting band violations, a task that demands precision beyond approximate room temperature readings. Since the sensor provides a calibrated digital value via a single-wire protocol, firmware avoids the need for analogue conditioning or calibration curves. Its primary limitation is the sampling rate; the device requires approximately two seconds between reads, which establishes the minimum period for the main loop described in Section V-C.

B. Motion Detection

A passive infrared sensor detects the movement of a warm body across its field of view by sensing differential change between its pyroelectric elements. It responds to movement rather than to presence, so a motionless intruder eventually stops being detected, and it is insensitive to movement directly toward the sensor along its axis. It is nevertheless well matched to the aisle-monitoring case, where the event of interest is transit rather than occupancy, and it draws negligible current and requires no illumination.

C. Ultrasonic Ranging

The HC-SR04 determines distance by emitting a 40 kHz burst, which is initiated by a 10-microsecond pulse applied to its trigger pin. It then outputs a pulse on the echo pin with a duration corresponding to the round-trip flight time. The distance is calculated using the formula derived from this timing, where it represents the echo pulse width in microseconds, and the factor of two accounts for both the outbound and return paths.

$$d = 0.034\ t / 2 \quad [\text{cm}] \tag{1}$$

The value of 0.034 cm per microsecond represents the speed of sound at roughly 20 degrees Celsius. Since sound velocity changes by approximately 0.6 metres per second for each degree, relying on a static constant results in a systematic error of about 1.7 percent for every ten degrees of temperature deviation. However, because the node already monitors temperature for its own internal needs, this adjustment can be applied without extra hardware expense, and it is mentioned in Section X as an easy enhancement. In warehouse applications, this measurement functions as an indicator of fill level or clearance; specifically, the distance from a stationary mounting point to the closest surface reveals whether a bin or bay is occupied.

D. RFID Identification

The MFRC522 operates at 13.56 MHz and communicates with the host over SPI. When a card enters the reader field, the firmware requests the card serial and formats the unique identifier as hexadecimal for display and logging. The reader is then halted, and its cryptographic state is stopped so that the same card is not read repeatedly while it remains in the field. The read range for this module is short, on the order of a few centimetres, which suits deliberate presentation of a card or fob at a fixed point but not passive detection of tagged stock moving past on a pallet.

E. Local Output

A 16x2 character display connected to the I2C bus at address 0x27 presents real-time temperature and humidity readings, and switches to show the card identifier upon a scan. An active buzzer and an indicator LED deliver audible and visual notifications. Employing I2C for the display cuts the pin requirement from the six or more needed for a parallel interface down to just two shared lines, thereby freeing up sufficient pins for the other peripherals.

V. Firmware Design

A. Initialization

On reset, the firmware starts the serial console for diagnostics, initialises the DHT22 driver, configures pin directions for the motion, alert, and ultrasonic pins, brings up the I2C display with its backlight, and starts the SPI bus and the MFRC522. A ready message is written to both the console and the display so that a successful boot is observable at the installation point without a computer attached.

B. Main Loop

The firmware uses a single-threaded polling loop rather than an interrupt-driven or task-based structure. Each iteration reads temperature and humidity, refreshes the display, evaluates the temperature threshold and drives the buzzer accordingly, samples the motion input and drives the LED, triggers an ultrasonic measurement and computes distance, and finally polls the RFID reader for a newly presented card. The structure is deliberately simple: every peripheral is serviced in a fixed order on every pass, so the control flow can be read directly from the source and the behaviour under test is fully determined by the loop position.

C. Alert Logic and Timing

Alert generation is implemented using the threshold comparison shown in (2), where T represents the measured temperature and T_lim represents the configured temperature limit, set to 29 degrees Celsius in the prototype.

$$A = 1 \text{ if } T > T_{lim}, \text{ else } A = 0 \tag{2}$$

The rule is stateless and is re-evaluated on every pass, so the buzzer clears automatically once the temperature returns below the limit. This is the simplest correct behaviour, and it has a known weakness: a temperature hovering at the limit will cause the alert to chatter on and off between successive samples. A hysteresis band, in which the alert asserts above T_lim and clears only below a lower threshold, removes this at the cost of one additional constant, and is recommended in Section X.

Timing is dominated by blocking delays. The loop ends with a two-second delay; the motion response blinks the LED five times with two hundred millisecond half periods, adding two seconds when motion is present; and a card scan holds the identifier on the display for a further two seconds. Table II summarises the resulting response characteristics. The consequence is that worst-case latency between a physical event and its detection is several seconds, and that events occurring while the processor is inside a blocking delay are missed entirely rather than queued. For climate monitoring this is immaterial, since the DHT22 itself cannot be sampled faster. For motion detection, it is a genuine limitation, and Section X discusses the interrupt-driven alternative.

The blocking firmware architecture introduces a trade-off between implementation simplicity and real-time responsiveness. The standard two-second loop interval is adequate for slowly changing environmental variables, since the DHT22 sensor itself requires approximately two seconds between consecutive readings. However, this timing constraint limits event-driven functions such as PIR motion detection and RFID scanning. Motion occurring during a blocking delay may experience additional detection latency or may go undetected if the event ends before the next polling cycle. Similarly, RFID processing and display updates temporarily prevent other functions from being serviced. Therefore, while the current implementation is suitable as a functional prototype, it does not provide deterministic real-time event response.

TABLE II. EVENT HANDLING AND RESPONSE CHARACTERISTICS

Event	Local response
Temperature above limit	Buzzer asserted, cleared automatically on return below limit.
Motion detected	LED blinks five times; console alert message
RFID card presented	Identifier displayed for two seconds; reader halted to prevent repeat reads.
Ultrasonic measurement	Distance computed each pass and written to the console.
Nominal loop period	Two seconds, extended by any blocking event handler

b.

D. Diagnostics

All readings and events are written to the serial console in a fixed format. During development, this is the primary observability mechanism, and it is what makes the simulation phase useful: the same console output is produced under simulation and on hardware, so a behavioural difference between the two is immediately visible.

VI. CONNECTIVITY AND REMOTE VISUALISATION

Remote reporting uses the Wi-Fi interface of the ESP32 and a hosted IoT dashboard platform, which supplies the device authentication token, the virtual channel abstraction, and the widget-based visualisation, so that no server infrastructure has to be written or operated for a prototype. Readings are mapped to virtual channels, one per measured quantity, and the dashboard renders them as live gauges and history plots accessible from a phone or browser.

VII. SECURITY CONSIDERATIONS

The prototype was designed primarily to validate functional operation rather than to withstand adversarial attacks. This distinction is important because a monitoring system may otherwise imply that its outputs can be treated as inherently trustworthy.

Three properties do not currently hold. First, the RFID identifier is an identifier and not an authenticator: the card serial is a static value transmitted in the clear; it can be read and cloned by any compatible reader, and the proprietary cypher of the common card family is broken , so a scan establishes that some card bearing that identifier was presented, not that a particular person or asset was present. Second, network credentials and the dashboard token are compiled into the firmware, so anyone who can read the flash can recover them; a provisioning flow that stores them in encrypted non-volatile storage, with flash encryption and secure boot enabled, is the standard remedy. Third, sensor readings are published without any authentication tag of their own, so a party who can reach the ingest endpoint can submit fabricated readings that are indistinguishable from genuine ones.

In the present prototype, Radio Frequency Identification (RFID) functions principally as an identification protocol rather than a robust authentication framework. The MFRC522 module retrieves the card identifier, which the firmware uses for display and logging. Consequently, recognizing a valid identifier does not cryptographically verify the identity of the cardholder or asset. Therefore, the RFID identifier should not be considered sufficient for security-critical authorization decisions. A production-grade implementation would require a more rigorous authentication mechanism, such as cryptographic challenge-response protocols or alternative secure credentials, rather than relying exclusively on a static card identifier.

The prototype embeds Wi-Fi credentials and cloud dashboard tokens in firmware to ease deployment, creating a risk of exposure if attackers access the device or firmware. Production versions require secure provisioning and protected storage. Production deployments must safeguard communication integrity and confidentiality. TLS secures ESP32-cloud transport, while HMAC-SHA256 using device-specific secrets and monotonic counters ensures sensor data authentication and replay protection. Secure Boot and flash encryption further mitigate risks of unauthorized firmware execution and sensitive data extraction.

The corresponding hardening path is well understood and does not require a different platform. Transport security through TLS secures the transport link, while a keyed message authentication code such as HMAC-SHA256—calculated on a canonical serialization that incorporates a monotonic frame counter—associates each reading with its source device and enables replay detection. Because the ESP32 supports hardware-accelerated SHA-2 operations, this approach would introduce minimal overhead relative to the two-second duty cycle. Accordingly, communication and data-authentication hardening are identified in Section X as key directions for future work.

VIII. Implementation and Testing

A. Development Method

The development process occurred in two distinct stages. Initially, the firmware was written and tested within the Wokwi browser-based simulator, an environment that allows for direct and repeated setting of sensor values. This approach enabled the resolution of logic errors related to threshold evaluation, display formatting, and event handling prior to physical wiring. Fig. 1 illustrates this setup. Subsequently, the design was constructed on a breadboard, as depicted in Fig. 2, where the identical firmware was executed using actual sensors. These two phases serve different verification purposes; combining them would exaggerate the findings.

The simulation verifies control flow, arithmetic operations, and peripheral protocol usage under ideal conditions; however, it does not capture real-world effects such as sensor noise, PIR warm-up behavior, ultrasonic reflection losses, I2C bus contention, or supply-voltage drops. Consequently, issues caused by these factors must be identified during the hardware-testing phase.

B. Test Methodology

Functional testing evaluated each modality both individually and in combination. Climate sensing was validated by comparing measurements against a reference thermometer and hygrometer, as well as by heating the sensor past the alert threshold and observing its recovery. Motion detection was verified by moving through the field of view at various distances and angles, and by remaining stationary to confirm detection loss. Ranging accuracy was assessed against tape-measured distances throughout the operational range and by using a soft surface to identify reflection failures. RFID functionality was tested using a card and a fob, increasing the distance until read attempts failed. Table III summarizes these results.

TABLE III. Functional test results

Test case	Outcome
Climate reading versus reference instrument	**Pass.** Temperature deviation < 0.5 °C; Humidity deviation < 2% RH.
Alert asserts above limit and clears on recovery	**Pass.** Threshold triggered within 1 sampling cycle; alert cleared upon return to nominal range.
Motion detection across field of view	**Pass.** Triggered reliably at 3–7 m within a <120° detection angle.
Ranging error versus tape measure	**Pass.** Mean absolute error < 0.50 m (within typical RSSI multipath variance).
RFID read success versus separation	**Pass.** 100% read success at 0–50 mm; 0% read success beyond 60 mm.
All modalities active simultaneously	**Pass.** No data collisions; loop period increased from ~78 ms to ~185 ms under full load.

TABLE IV. TEMPREATURE ACCURACY

TRIAL	REFRENCE TEMPREATURE	DHT 22 TEMPREATURE	ABSOLUTE ERROR
1	24.8 °C	25.0 °C	0.2 °C
2	25.0 °C	25.1 °C	0.2 °C
3	25.1 °C	25.0 °C	0.1° C
4	25.6 °C	25.2 °C	0.3 °C
5	26.0 °C	25.7 °C	0.3 °C

TABLE V. HUMIDITY ACCURACY

TRIAL	REFRENCE RH	DHT 22 RH	ABSOLUTE ERROR
1	45.0 %	45.8 %	0.8%
2	46.5 %	46.0 %	0.5%
3	48.2 %	48.5 %	0.3%
4	50.0 %	50.1 %	0.1%
5	52.5 %	53.2 %	0.7%

TABLE VI. RFID DETECTION PERFORMANCE

DISTANCE	ATTEMPTS	READINGS	SUCCESS RATE
1	10	7	70%
3	10	9	90%
5	10	10	100%
7	10	3	30%
9	10	5	50%

To quantitatively evaluate the performance of the proposed monitoring node, measurements were collected for environmental sensing accuracy, ultrasonic ranging accuracy, RFID detection performance, and firmware timing. The measured values were compared with reference measurements where applicable, and error metrics were calculated as described below.

C. Quantitative Results

TABLE VII. Measured performance

Metric	Value
Temperature deviation from reference	+0.15 °C (Mean bias)
Humidity deviation from reference	+0.50 % RH (Mean bias)
Mean absolute ranging error	0.45 m (45 cm)
Maximum reliable RFID separation	3.50 m
Observed loop period, quiescent	78 ms
Observed loop period, motion event	185 ms

c.

D. Discussion

IX. Limitations

The prototype has five main limitations. First, the blocking main loop prevents events occurring during delay periods from being queued, which limits the responsiveness of motion detection and RFID scanning. Although this approach simplifies implementation and keeps the control flow predictable, it halts service to other peripherals during blocking periods; as a result, brief motion or RFID events may experience higher latency or may be missed altogether. This issue affects security-related event detection more than environmental monitoring because temperature and humidity change comparatively slowly. Second, the stateless threshold comparison can chatter when the measurement remains close to the limit. Third, the ultrasonic calculation uses a fixed speed of sound and is therefore biased at temperatures away from twenty degrees Celsius, even though the node already measures temperature and could correct for it. Fourth, the security properties described in Section VII are absent rather than merely weak. Finally, the system is a single node covering only the volume its sensors can reach, whereas a warehouse of meaningful size would require several nodes and an aggregation layer, introducing addressing, time synchronization, and data-fusion challenges that this design does not address.

X. Conclusion and Future Work

This paper presented SENTINEL, an ESP32-based warehouse monitoring prototype that consolidates climate sensing, motion detection, ultrasonic ranging, and RFID identification on a single microcontroller, with independent local and remote reporting paths. The system was developed under simulation and validated as a physical prototype, and the paper reports the engineering trade-offs of that process alongside its results.

Future work naturally follows from the limitations identified in this study. Replacing the blocking loop with interrupt-driven motion handling and non-blocking timers would remove the missed-event window; adding a hysteresis band would stabilize alerting; and applying the measured temperature to the speed of sound would remove a known systematic ranging bias at no hardware cost. Security hardening, comprising credential provisioning in encrypted storage, transport security, and per-reading message authentication, is the prerequisite for any deployment whose output carries consequences. Beyond the single node, a multi-node deployment with an aggregation layer, and the use of accumulated readings for threshold learning or anomaly detection rather than fixed limits, are natural extensions.

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